

Sr. No.	Title	CO	Marks	Hours
4	Write and Perform the SQL queries to implement different operators(i.e. +,-,/,*,%) and SQL Aggregate Functions. (Assume the necessary database OR use the database of Practical-1:) Write a SQL statement to find the total purchase amount of all orders. Write a SQL statement to find the average purchase amount of all orders. Write a SQL statement to find the number of salesmen currently listing for all their customers. Write a SQL statement to know how many customers have listed their names. Write a SQL statement to find the number of customers who get at least a rating for his/her performance. Write a SQL statement to know the maximum and minimum purchase amount of all the orders. Write a SQL statement that selects the highest grade for each of the cities of the customers. Write a SQL statement to find the highest purchase amount ordered by each customer with their ID and highest purchase amount. Write a SQL statement to find the highest purchase amount ordered by each customer on a particular date with their ID, order date, and highest purchase amount. Write a SQL statement to find the highest purchase amount on the date '01-JAN-06' for each salesman with their ID Write a SQL statement to find the highest purchase amount with their ID and order date, for only those customers who have the highest purchase amount in a day is more than 100	2	10	4

Write and Perform the SQL queries to implement different operators (i.e. +,-,/,*,%) and SQL Aggregate Functions. (Assume the necessary database OR use the database of Practical-1:)

SQL> SELECT 15+10 FROM dual;

```

15+10
-----
25
=====

```

SQL> select * from customer;

```

CNUM CNAME          CITY          RATING  SNUM
-----
201 Aladin          Goa          1      101

```

202 Malaram	Surat	1	101
203 Ram	Surat	2	102
204 Kishan	Bharuch	3	104
205 Mehul	Dhandhinagar	4	105

SQL> select * from orders;

ONUM	AMT	ODATE	CNUM	SNUM
-----	-----	-----	-----	-----
301	100	01-JAN-06	201	101
302	120	15-FEB-10	202	102
303	130	14-FEB-04	203	103
304	90	15-FEB-03	204	104
305	89	16-APR-00	205	105

SELECT cname, rating, **orders.amt*0.10**
FROM customer, orders
WHERE orders.cnum=customer.cnum;

SQL> SELECT cname, rating, orders.amt-0.10
2 FROM customer, orders
3 WHERE orders.cnum=customer.cnum;

CNAME	RATING	ORDERS.AMT-0.10
-----	-----	-----
Aladin	1	99.9
Malaram	1	119.9
Ram	2	129.9
Kishan	3	89.9
Mehul	4	88.9

1. Write a SQL statement to find the total purchase amount of all orders

SQL> SELECT SUM (amt)
2 FROM orders;

SUM(AMT)

529

2. Write a SQL statement to find the average purchase amount of all orders.

SQL> SELECT AVG (amt)
2 FROM orders;

AVG(AMT)

105.8

3. Write a SQL statement to find the number of salesman currently listed for all of their customers.

```
SQL> SELECT COUNT (DISTINCT snum)
2 FROM orders;
```

COUNT(DISTINCT SNUM)

```
-----
5
```

4. Write a SQL statement to know how many customers have listed their names.

```
SQL> SELECT COUNT(*)
2 FROM customer;
```

COUNT(*)

```
-----
5
```

5. Write a SQL statement to find the number of customers who gets at least a rating for his/her performance.

```
SELECT COUNT(DISTINCT cnum) AS number_of_customers
FROM customer
WHERE rating IS NOT NULL;
```

```
SQL> select * from customer;
```

CNUM	CNAME	CITY	RATING	SNUM
201	Aladin	Goa	1	101
202	Malaram	Surat	1	101
203	Ram	Surat	2	102
204	Kishan	Bharuch	3	104
205	Mehul	Dhandhinagar	4	105

```
SQL> SELECT COUNT (ALL rating)
2 FROM customer;
```

COUNT(ALLRATING)

```
-----
5
```

6. Write a SQL statement to know the maximum and minimum purchase amount of all the orders.

```
SELECT
MAX(amt) AS max_purchase_amount,
MIN(amt) AS min_purchase_amount
FROM orders;
```

```
SQL> select * from orders;
```

ONUM	AMT	ODATE	CNUM	SNUM
------	-----	-------	------	------

301	100	01-JAN-06	201	101
302	120	15-FEB-10	202	102
303	130	14-FEB-04	203	103
304	90	15-FEB-03	204	104
305	89	16-APR-00	205	105

SQL> SELECT MAX (amt) from orders;

```
MAX(AMT)
-----
130
```

SQL> SELECT Min (amt) from orders;

```
MIN(AMT)
-----
89
=====
```

7. Write a SQL statement that selects the highest grade/rating for each of the cities of the customers.

SQL> SELECT city,MAX(rating)
2 FROM customer
3 GROUP BY city;

```
CITY          MAX(RATING)
-----
Gandhinagar    4
Goa            1
Bharuch        3
Surat          2
=====
```

8. Write a SQL statement to find the highest purchase amount ordered by each customer with their ID and highest purchase amount.

SQL> SELECT cnum,MAX(amt)
2 FROM orders
3 GROUP BY cnum;

```
CNUM  MAX(AMT)
-----
205    89
201   100
202   120
203   130
204    90
=====
```

9. Write a SQL statement to find the highest purchase amount ordered by each customer on a particular date.

```
SQL> SELECT cnum,orders.odate,MAX(amt)
2 FROM orders
3 GROUP BY cnum, date;
```

CNUM	ODATE	MAX(AMT)
205	16-APR-00	89
201	01-JAN-06	100
203	14-FEB-04	130
202	15-FEB-10	120
204	15-FEB-03	90

10. Write a SQL statement to find the highest purchase amount on a date '01-JAN-06' for each salesman with their ID

```
SQL> SELECT snum,MAX(amt)
2 FROM orders
3 WHERE odate = '01-JAN-06'
4 GROUP BY snum;
```

SNUM	MAX(AMT)
101	100

Write a SQL statement to find the highest purchase amount with their ID and order date, for only those customers who have the highest purchase amount in a day is more than 100

```
SQL> SELECT cnum,orders.odate,MAX(amt)
2 FROM orders
3 GROUP BY cnum,odate
4 HAVING MAX(amt)>100;
```

(Note:

1. The HAVING clause is used instead of WHERE with aggregate functions. While the GROUP BY Clause groups rows that have the same values into summary rows. The having clause is always used after the group By clause.

2. An aggregate function can be used in a WHERE clause only if that clause is part of a subquery of a HAVING clause and the column name specified in the expression is a correlated reference to a group.

)

CNUM ODATE	MAX(AMT)

203 14-FEB-04	130
202 15-FEB-10	120